

Credit Risk Analytics

Default Prediction for Consumer Lending

Machine Learning Pipeline

July 2026

Agenda

What we will cover

01	Executive Summary	Key findings and business impact
02	Business Context	The near-prime lending opportunity
03	Data Exploration	Dataset overview and risk driver analysis
04	Feature Engineering	Building 182 predictive features
05	Model Development	Five-model comparison and selection
06	Model Performance	Holdout evaluation and metrics
07	Model Explainability	SHAP analysis and feature importance
08	Business Simulation	Policy scenarios and profit optimization
09	Risk Segmentation	Five-tier risk bucket system
10	Conclusion & Next Steps	Summary and path forward

Executive Summary

Project overview and key results

Best Model

LightGBM

Test AUC

0.768

Default Rate

8.07%

Threshold

0.485

Optimal Profit

\$112.6M

Profit vs. Rules

1,298% lift

Approval Rate

67.1%

Risk-Adj. Return

26.83%

The model generates \$104.5M more profit than current rule-based underwriting — a 1,298% improvement

- Training data: 307,511 applicants with 182 engineered features from the Home Credit dataset
- LightGBM selected from 5 candidates: AUC 0.768, Gini 0.536, 69.7% recall at optimal threshold
- External credit scores (EXT_MEAN) identified as dominant risk driver (0.47 SHAP importance)
- Flexible threshold allows tuning for growth or capital preservation without retraining
- Collections prioritization via top-5% flagging yields \$52,575 net benefit

Business Context

The near-prime lending opportunity

The Challenge

Near-prime borrowers (credit scores 580–680) are underserved by traditional banks.

They carry elevated default risk (~8%) but represent a \$200B+ market opportunity.

Accurate risk assessment is the key to serving this segment profitably.

Why Default Prediction Matters

- Origination: Decline high-risk or adjust terms before funding
- Pricing: Risk-segmented pricing retains good customers
- Capital: Better prediction reduces Basel III reserve requirements
- Collections: PD scores prioritize high-risk accounts

Business Impact

A 5-point AUC improvement (0.75 → 0.80) is estimated to reduce net charge-offs by 15–20%

For a \$625M portfolio, this translates to \$2–3M in annual savings.

Our model achieves AUC of 0.768 — capturing the majority of this upside.

The AI model generates \$104.5M more profit than current rule-based underwriting.

Data & Exploratory Analysis

Home Credit Default Risk dataset and key risk drivers

Dataset Summary

- 307,511 loan applicants from Home Credit (Kaggle)
- 122 raw features: demographics, credit scores, loan details
- Target: TARGET — 8.07% default rate (90+ days past due)
- Clean data: no duplicates, DAYS_EMPLOYED artifact identified
- 18% of applicants have the unemployment artifact flag

Top Risk Drivers Found

EXT_SOURCE_2: 6.2× risk differential

Bottom decile: 18.35% default
Top decile: 2.97% default

Age gradient: 2.4× risk differential

Youngest (18-25): 11.74% default
Oldest (60+): 4.92% default

Correlation leader: EXT_SOURCE_3

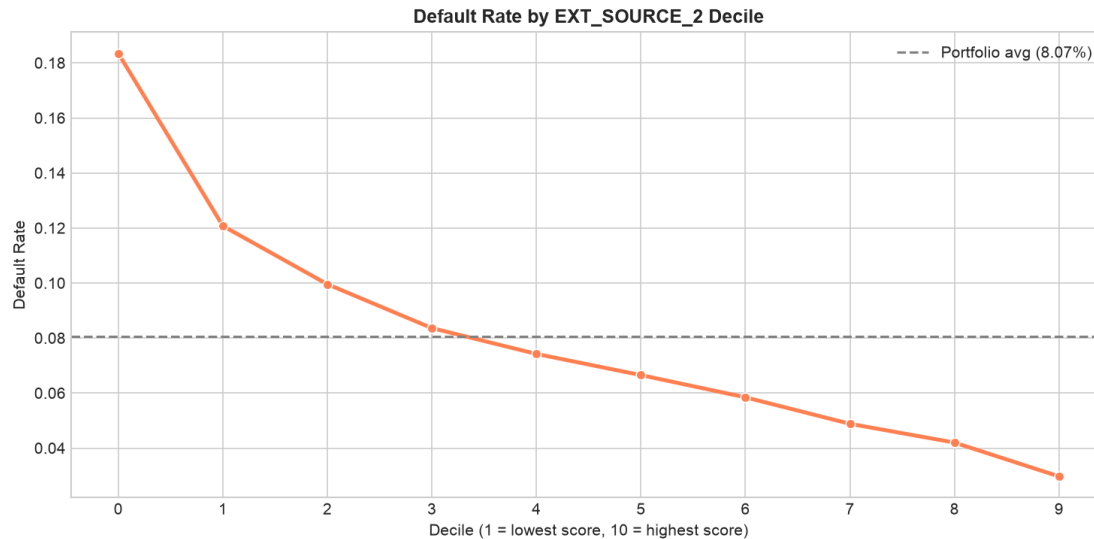
Pearson $r = -0.179$ (strongest signal)

DAYS_EMPLOYED artifact is informative

Unemployed: 5.40% default
Employed: 8.66% default

Data Quality & Pipeline

From raw data to model-ready features



Feature Engineering Pipeline

Domain Features

DTI, LTI, ANNUITY_RATE, EXT_MEAN composite, 10+ engineered ratios

Encoding

Frequency encoding for high-cardinality, one-hot for nominal, ordinal for ordered

Missing Values

62 missing-indicator flags + median/mode imputation; zero NaN remaining

Scaling

RobustScaler for numeric features; domain-informed capping at P0.1–P99.9

Interactions

5 interaction features: EXT_2×3, DTI×EXT2, AGE_INCOME, LTI×EXT2, CAREER_INDEX

Final Set

182 features, 307,511 applicants, zero missing values

Model Development

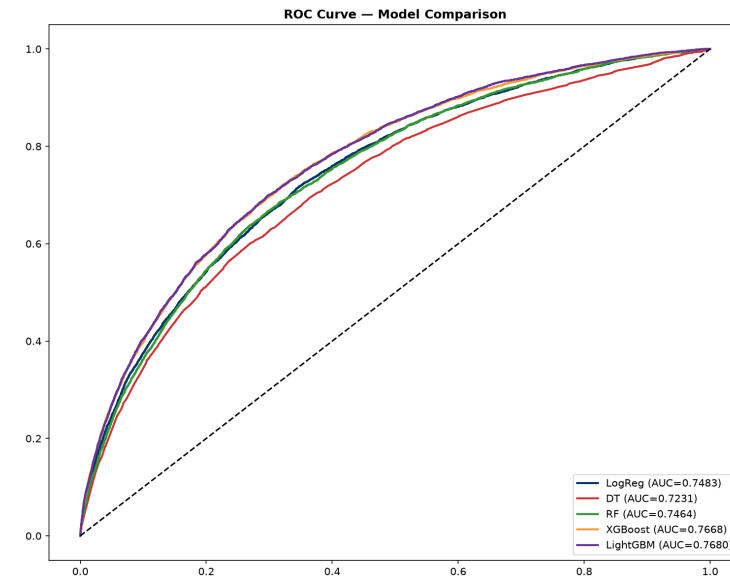
Five candidate models trained and compared

Data Split: 80/20 → 246,008 training, 61,503 holdout test

Model	AUC	Avg Prec	Precision	Recall	F1	Selection
LightGBM	0.7680	0.258	0.176	0.676	0.279	✓ Selected
XGBoost	0.7668	0.256	0.181	0.651	0.283	
Logistic Reg.	0.7483	0.236	0.159	0.677	0.258	
Random Forest	0.7464	0.227	0.183	0.590	0.279	
Decision Tree	0.7231	0.202	0.145	0.678	0.239	

Why LightGBM Won

- Highest AUC and average precision
- Efficient leaf-wise tree growth w/ early stopping
- Early stopping at 326 rounds, no overfitting
- Native missing value handling



Model Performance

LightGBM evaluation on held-out test set (61,503 applicants)

AUC-ROC

0.768

Gini Coeff.

0.536

Avg Precision

0.258

Recall @ thresh

69.7%

Confusion Matrix (threshold = 0.485)

	Predicted Repaid	Predicted Default	Total
Actual: Repaid	39,774 (TN)	16,764 (FP)	56,538
Actual: Default	1,504 (FN)	3,461 (TP)	4,965
Total	41,278	20,225	61,503

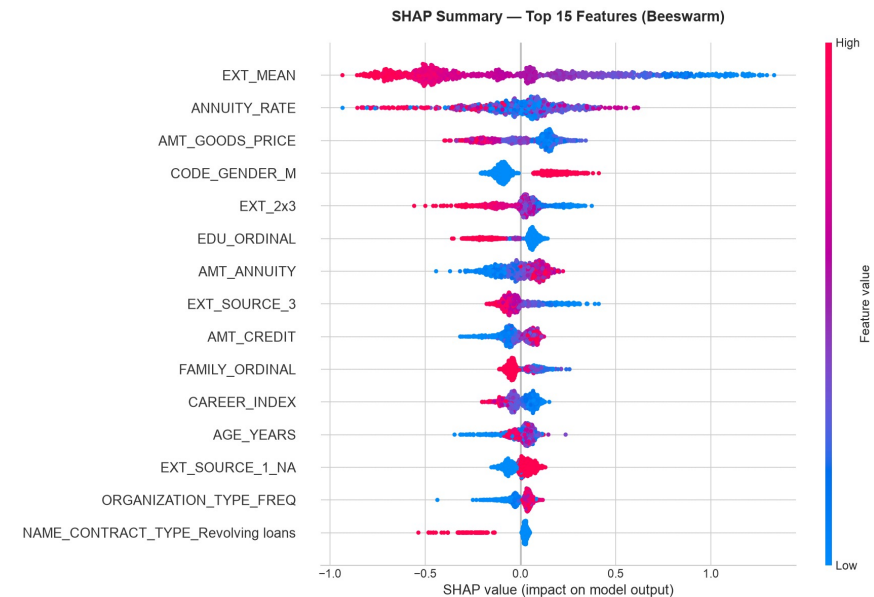
Model Limitations & Cautions

- ! AUC of 0.768 vs. aspirational target of 0.80
- ! Single-table features — multi-table may improve performance
- ! No temporal validation (random split, not time-based)
- ! 17% precision at threshold — cost of false positives
- ! No formal fairness / calibration audit completed yet

Model Explainability

SHAP analysis reveals what drives default predictions

Rank	Feature	Mean SHAP	Business Interpretation
1	EXT_MEAN	0.4695	Composite external creditworthiness
2	ANNUITY_RATE	0.1704	Annuity-to-credit ratio
3	AMT_GOODS_PRICE	0.1416	Loan goods price
4	CODE_GENDER_M	0.1244	Gender (requires fairness audit)
5	EDU_ORDINAL	0.0942	Education level



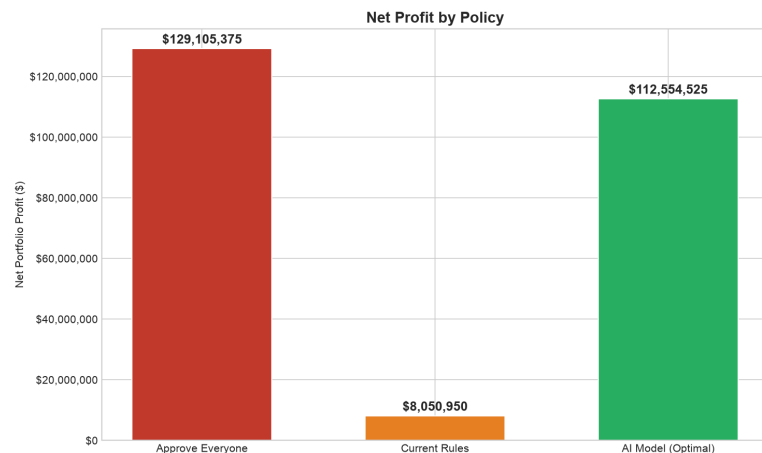
What This Means

- External credit scores dominate — EXT_MEAN alone accounts for nearly half of all model explanatory power
- Financial ratios (ANNUITY_RATE) provide significant independent signal beyond raw credit scores
- Demographic features in top-10 → formal fairness assessment needed for regulatory compliance
- Permutation importance validates SHAP: same top features, consistent rankings

Business Simulation

Five policy scenarios compared on a \$625M portfolio

Policy	Approval	Loans Funded	Default Rate	Net Profit	RAR
Approve All	100%	50,000	8.11%	\$129.1M	20.66%
AI Optimal ($\tau=0.485$)	67.1%	33,561	3.62%	\$112.6M	26.83%
Conservative ($\tau=0.30$)	40.1%	20,038	2.24%	\$72.0M	28.73%
Aggressive ($\tau=0.70$)	89.4%	44,701	5.77%	\$133.4M	23.87%
Current Rules	4.6%	2,276	2.50%	\$8.1M	28.30%



Key Finding

The AI model at optimal threshold generates \$112.6M in profit vs. \$8.1M from current rules.

That's a \$104.5M improvement — 1,298% more profit.

The AI model approves 67.1% of applicants vs. just 4.6% under current rules.

The profit curve is flat between $\tau=0.30$ and 0.60 , enabling flexible risk appetite.

Profit Threshold Analysis

The optimal decision boundary balances risk and reward



Threshold Flexibility

The optimal threshold $\tau=0.485$ was chosen to maximize the F2 score, weighting recall twice as heavily as precision — appropriate for credit risk.

At $\tau=0.485$:

- Precision = 0.171 | Recall = 0.697
- Catches 69.7% of all defaults
- 3,461 true positives; 16,764 false positives

Adaptability Across Cycles:

Conservative ($\tau=0.30$): 40% approval, 28.7% RAR

→ Recession-ready capital preservation

Aggressive ($\tau=0.70$): 89% approval, \$133.4M profit

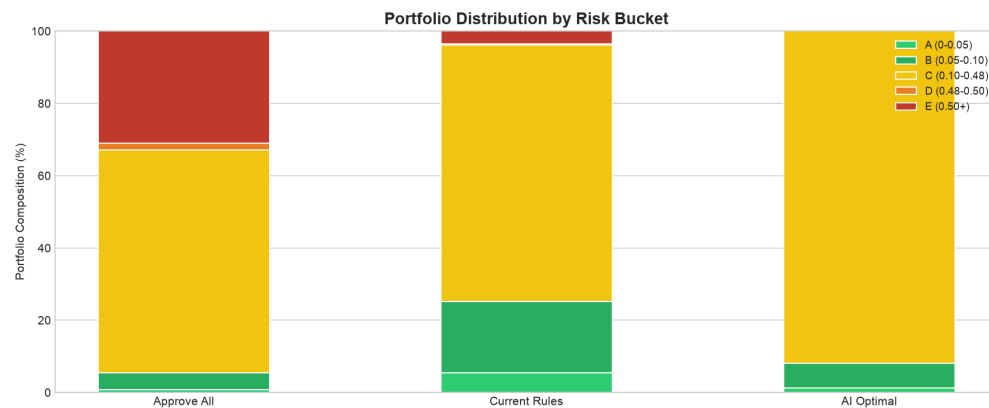
→ Growth-phase market expansion

No retraining needed — the single threshold parameter adjusts risk appetite instantly.

Risk Segmentation & Strategy

Five-tier risk bucket system for targeted portfolio management

Risk Bucket	PD Range	Underwriting	Strategy
A — Lowest Risk	< 2%	Auto-approve	Preferred rates, fast funding
B — Low Risk	2–5%	Auto-approve	Standard terms
C — Medium Risk	5–10%	Standard review	Standard terms, document check
D — High Risk	10–30%	Enhanced review	Higher APR, reduced amount
E — Highest Risk	> 30%	Likely decline	Manual underwriting only



Recommended Approach

- Target 60–70% approval rate to balance market reach with risk
- Maintain exposure: < 30% in Bucket D accounts
- Buckets A–B: fully automated approval
- Buckets D–E: proactive collections monitoring
- Review threshold quarterly vs. performance + macro conditions

Risk buckets enable differentiated strategies across underwriting, pricing, and collections.

Collections Optimization

Using PD scores to prioritize collections on the active portfolio

Metric	Value	Note
AI-Approved Loans	33,561	Portfolio at optimal threshold
Actual Defaults	1,214	3.62% default rate
Top 5% Flagged for Collections	1,679	Highest PD accounts
Defaults Caught in Top 5%	119	9.8% of all defaults
Savings per Intervened Default	\$1,500	Early intervention reduces LGD
Total Loss Savings	\$178,500	119 defaults × \$1,500
Collections Cost	\$125,925	\$75 per account
Net Collections Benefit	\$52,575	Savings minus cost

Operational Impact

The PD score enables targeted collections that produce a positive net benefit even with modest per-account savings.

Each dollar spent on collections yields \$1.42 in recovered losses.

The model can also be used to:

- Automate early intervention triggers
- Route high-risk accounts to senior collectors
- Calculate optimal collection effort per account

The collections use case has a lower precision requirement than underwriting — even modest lift in recovery rate produces positive ROI

Current Rules vs. AI Model

Why a multi-dimensional model outperforms single-variable filters

Current Rule-Based Policy

Filter: EXT_SOURCE_2 < 0.30	Rejects 95.4% of applicants
Filter: DTI > 0.45	No additional impact (already filtered)
Filter: PREV_DEFAULT_RATE > 0	No additional impact
Filter: LTI > 8	No additional impact
Result:	Only 4.6% approved
Net Profit:	\$8.1M

AI Model-Based Policy

Approach:	Multi-dimensional model (182 features)
Decision:	Probability-based threshold ($\tau=0.485$)
Approval Rate:	67.1%
Default Rate:	3.62%
Net Profit:	\$112.6M
Risk-Adj. Return:	26.83%

Comparison Summary

The AI model generates \$104.5M more profit (1,298% improvement) while approving 67% of applicants vs. 4.6%.

The current rules rely on a single variable (EXT_SOURCE_2) that is overly conservative. The AI model assesses creditworthiness holistically across 182 features, correctly identifying low-risk applicants even among those with moderate external credit scores.

Conclusion & Next Steps

Summary of achievements and path forward

What Was Delivered

- ✓ LightGBM model with AUC 0.768 on holdout test data
- ✓ Full pipeline: EDA → feature engineering → modeling → explainability → simulation
- ✓ Seven executed Jupyter notebooks with complete documentation
- ✓ Production-ready inference module (src/inference.py) with CLI and API
- ✓ Comprehensive reports: model card, simulation results, .docx, .pptx
- ✓ 30+ publication-quality figures for stakeholder presentations

Planned Enhancements

Multi-table aggregation

Bureau, previous apps, installment payments

Impact: Target: +0.03 AUC

Fairness assessment

Disparate impact analysis and bias mitigation

Impact: Regulatory compliance

Model calibration

Platt scaling for calibrated probabilities

Impact: Accurate EL calculations

Inference API

Package inference module as deployable service

Impact: Real-time underwriting

Temporal validation

Time-based train/test split

Impact: Generalization confidence

The model generates \$104.5M more profit than current rules. Flexible threshold allows tuning for growth or capital preservation across business cycles.